End Semester Examination, April – May, 2025

Degree: **BCA** Semester: **Fourth**

Stream: **BCA**

Paper Code: **BCACC402**

Paper Name: **Programming with JAVA**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. State whether the following statements are true or false **2**
   a. The main emphasis of procedure-oriented programming is on algorithms rather than on data.
   b. One of the striking features of object-oriented programming is the division of programs into objects that represent real-world entities.
   c. Wrapping up of data of different types into a single unit is known as encapsulation.
   d. One problem with OOP is that once a class is created, it can never be changed.

2. How does Java's security model compare to other programming languages in preventing vulnerabilities? **2**

3. List at least five major C++ features that were intentionally removed from Java? **2**

4. What is the task of the **"main"** method in a Java program? **2**

5. What are the three parts of a simple, empty class? **2**

6. How do we invoke a constructor? **2**

7. Correct the error in the following code. **2**

```
class VarName {
public static void main(String[] args) {
System.out.println("Hello World!");
}
}
```

8. Write a statement to declare and instantiate an array to hold marks obtained by students in different subjects in a class. Assume that there are up to 60 students in a class and there 8 subjects. 2

9. What is the role of the catch block when used alongside a try block in Java? 2

10. The code throws an exception in case the desired input is not received. Debug the code 2

```
class excep3 {
public static void main(String args[]) {
if (args[0].equals("Hello")
System.out.println("String is right");
else
throw new Exception("Invalid String");
}
}
```

11. Is it necessary to catch every type of exception in a Java program? 2

12. Debug the following code for reading a file using FileInputStream class. 2

```
import java.io.*;

class file2 {
public static void main(String args[]) {
if (args.length == 1) {
try {
FileInputStream fstream = new FileInputStream(args[0]);
DataInputStream in = new DataInputStream();
while (in.available() > 0) {
System.out.println(in.readLine());
}
in.close();
} catch (Exception e)
{
System.err.println("File Input Error ");
}
}
else {
System.out.println("Invalid parameters");
}
}
}
```

13. What is an applet? 2

14. Is there a limit to the number of <PARAM> tags that can be used in an applet? 2

15. What is multi-threaded programing? 2

Part - B

Attempt any 6 questions out of 9 questions

Each question carries 5 Marks (5 × 6)

16. Explain the concept of garbage collection in JAVA 5

17. Develop a Java program to get a number from parameterized constructor and print whether it is positive or negative. 5

18. How do multiprocessing and multithreading differ in terms of resource utilization, and how can they be implemented in a program? 5

19. Explain how Java handle memory allocation for String objects created using string literals? 5

20. What is the process for including a class or an interface in a specific package in Java? 5

21. Create a try block that is likely to generate three types of exception and then incorporate necessary catch blocks to catch and handle them appropriately. 5

22. Which exception may be thrown if the given code is executed? 5

```
class excep2 {
public static void main(String args[]) {
try {
int n = Integer.parseInt(args[0]);
int n1 = Integer.parseInt(args[1]);
int n2 = n + n1;
System.out.println("Sum is " + n2);
} catch(ArithmeticException ex) {
System.out.println("Arithmetic Exception : " + ex.getMessage());
} catch (NumberFormatException ex) {
System.out.println("Format Exception : " + ex.getMessage());
} catch (Exception ex) {
System.out.println("Exception 2  : " + ex);
}
}
}
```

23. Why are interfaces used for achieving abstraction in Java? 5

24. Write and explain the syntax of inheritance of interfaces. 5

Part - C

Attempt any 5 questions out of 8 questions

Each question carries 10 Marks (10 × 5)

25. Define a class Library with attributes bookId, title, and author. Create an array of objects to store book information. Implement methods to search for a book by title and display all books by a specific author. **10**

26. Create a superclass Shape with a method area () and subclasses Circle, Rectangle, and Triangle that override the area method. Use an array of Shape objects to calculate and display the areas of different shapes. **10**

27. How do state transitions occur in the lifecycle of a thread, and what factors influence these transitions in Java? **10**

28. Create an abstract class BankAccount with abstract methods deposit() and withdraw(). Create two subclasses SavingsAccount and CurrentAccount that implement these methods. Write a Java program to manage multiple accounts using an array of objects. **10**

29. Write a program in java to create Box class with parameterized constructor with an object argument to initialize length, breadth and height also create a function volume which returns the volume of the box and print it in main method. **10**

30. How does the throw keyword differ from throws in Java, and how can custom exceptions be implemented using these keywords? Write a Java program that creates a method to check if a number is negative and throws an IllegalArgumentException if it is. **10**

31. Explain the different stages of an applet's life cycle interact, and what role does each stage play in its execution? **10**

32. Write a Java program to create a class Student with attributes name, rollNumber, and marks. Use ObjectOutputStream to serialize an object of this class and save it to a file. Then, use ObjectInputStream to deserialize the object and display its data. Explain why serialization is important in Java stream **10**

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